

Project 02

AI Music & Audio Analyzer

Living Project Documentation — v0.1

Item	v0.1 Baseline
Project Status	In Progress
Platform	Windows Laptop
Programming Language	Python 3.14.7
Development Environment	Visual Studio Code + PowerShell
Python Environment	Virtual environment (venv)
Primary Hardware	Windows Laptop + Airdopes Joy Bluetooth Earbuds
Audio Input	Airdopes Joy Hands-Free microphone
Audio Output	Airdopes Joy Stereo
Python Audio Library	sounddevice
Initial Recording	5-second Airdopes microphone test
Recording File	audio\airdopes_test.wav

1. Project Definition

AI Music & Audio Analyzer is a Python-based learning and development project focused on capturing, processing, visualizing, and analyzing audio using a Windows laptop and Bluetooth earbuds. The project will progress from basic audio capture and signal processing toward music features, AI interpretation, and a user dashboard.

The new documentation starts at v0.1. Previous Project-02 reports and their version history are intentionally not used as the baseline for this report.

2. Project Objective

Build an audio-analysis application that can use the Airdopes Joy microphone as a real hardware audio input, process captured audio with Python, extract useful signal and music characteristics, visualize the results, and later add AI-based interpretation and Bluetooth playback.

3. Hardware Setup

Hardware	Purpose	Status
Windows Laptop	Python development, processing and dashboard	Available
Airdopes Joy Bluetooth Earbuds	Microphone input and audio playback	Connected / Detected
Internet Connection	Required later for cloud AI features	Available when required

4. Software Setup

Python 3.14.7 is being used in a project-specific virtual environment. Development is performed in Visual Studio Code with PowerShell. The sounddevice package is being used for Windows audio-device access and microphone capture.

5. Initial Hardware Architecture

Airdopes Joy Microphone → Bluetooth → Windows Audio System → Python sounddevice → NumPy / Audio Processing → Analysis → Dashboard / Report

Airdopes Joy Stereo is available as the playback path and will be integrated in a later stage.

6. Step 1 — Audio Device Detection

Status: COMPLETED

Python audio-device enumeration successfully detected the Windows audio endpoints. The Airdopes Joy Hands-Free microphone was identified as device ID 2 with one input channel and a reported sample rate of 44,100 Hz. Airdopes Joy Stereo was identified as a two-channel, 44,100 Hz output device.

Device	ID	Input	Output	Rate
Airdopes Joy Hands-Free	2	1	0	44.1 kHz
Airdopes Joy Stereo	4	0	2	44.1 kHz
Microphone Array (Realtek)	1	2	0	44.1 kHz
Speakers / Headphones (Realtek)	6	0	2	44.1 kHz

7. Step 2 — Airdopes Microphone Verification

Status: COMPLETED

The Airdopes Joy Hands-Free microphone was selected from Python using sounddevice. A microphone test produced a non-zero audio signal, confirming that the Bluetooth microphone is accessible to the Python application.

8. Step 3 — Airdopes Audio Recording

Status: COMPLETED

A five-second recording was captured from Airdopes Joy microphone device ID 2 and saved successfully as a WAV file.

Parameter	Actual Result
Device ID	2
Device	Headset (Airdopes Joy Hands-Free)
Input Channels	1
Sample Rate	44,100 Hz
Duration	5.00 seconds
Samples	220,500
RMS Amplitude	0.031082
Peak Amplitude	0.423889
Output File	audio\airdopes_test.wav

The sample count is consistent with the recording configuration: $44,100 \text{ samples/second} \times 5 \text{ seconds} = 220,500$ samples. The non-zero RMS and peak measurements confirm that an audio signal was captured.

9. Current Development Status

Stage	Description	Status
1	Project definition and hardware selection	Completed
2	Python / venv / VS Code / PowerShell setup	Completed
3	Windows audio device detection	Completed
4	Airdopes microphone verification	Completed
5	Five-second WAV recording	Completed
6	WAV loading and basic analysis	Next
7	Waveform visualization	Pending
8	RMS / Peak / dBFS analysis	Pending
9	FFT / frequency spectrum	Pending
10	Spectrogram	Pending
11	BPM / beat analysis	Pending
12	Music feature extraction	Pending
13	AI analysis	Pending

14	Streamlit dashboard	Pending
15	Airdopes stereo playback integration	Pending
16	Real-time analyzer	Planned

10. Initial Project Structure

Project-02-AI-Music-Audio-Analyzer/

■■■ audio/ → recorded and test audio files

■■■ reports/ → analysis graphs and reports

■■■ src/ → Python source code

■■■ tests/ → project tests

■■■ venv/ → Python virtual environment

■■■ requirements.txt

■■■ README.md

11. Testing Record

Test	Result
Audio device enumeration	PASS
Airdopes Joy microphone detected	PASS
Airdopes Joy Stereo output detected	PASS
Airdopes microphone capture	PASS
5-second WAV generation	PASS
WAV audio analysis	NEXT
Waveform generation	PENDING
FFT analysis	PENDING

12. Version History

v0.1 — 19-Aug-2026: New documentation baseline created from the beginning. Previous Project-02 report versions are excluded from this version history. Project definition, Windows/Airdopes hardware architecture, audio-device detection, Airdopes microphone verification, and five-second recording test are documented as the first completed milestones.

13. Next Step

Step 4 — Load the captured airdopes_test.wav file with Librosa, inspect duration/sample rate/samples, and begin waveform visualization. The report will be updated again only after the actual Step 4 test is completed.

This is a living project report. Future results will be added only after they are actually tested.